

1) A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.

Step	Troubleshooting Action	What I am Checking For
1. Check the power outlet and adapter	Connect the charger to a known-working wall outlet and inspect the charger, cable, and connector.	Checking whether the outlet, AC adapter, or charging cable is faulty or damaged.
2. Check charging/power indicators	Connect the charger and observe the charging LED or other power indicators.	Checking whether the laptop is receiving power and whether the battery/charging circuit responds.
3. Perform a power reset	Disconnect the charger, remove the battery if removable, then hold the power button for about 15–30 seconds. Reconnect the charger and try powering on.	Clearing residual electrical charge and checking whether the laptop starts after a power-state reset.
4. Check the battery and charging connection	If possible, test with the battery disconnected or use a known-good compatible charger.	Determining whether the battery, charging port, or power adapter is preventing the laptop from starting.
5. Check internal hardware	If there is still no response, inspect internal components such as RAM, motherboard connections, and signs of liquid or physical damage.	Checking for loose RAM, damaged components, short circuits, or motherboard/power-circuit failure.

2) You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.
Hint : Consider checking both hardware connections and basic display settings.

1. Confirm the symptoms

- Check that the PC is powered on and fans/LEDs are running.
- Confirm whether the monitor displays “No Signal”, remains completely black, or shows an error message.

2. Check the monitor power

- Make sure the monitor is switched on.
- Check the power cable and power outlet.
- Confirm that the monitor's power LED is illuminated.

3. Check the display cable

- Check the HDMI, DisplayPort, DVI, or VGA cable at both ends.
- Disconnect and reconnect the cable securely.
- If available, test with a known-good cable.

4. Check the correct input source

- Use the monitor's menu buttons to select the correct input, such as HDMI or DisplayPort.
- Make sure the cable is connected to the selected input.

5. Check the graphics output

- If the PC has a dedicated GPU, make sure the monitor cable is connected to the GPU's video output, not the motherboard output.
- Check that the GPU is properly seated and its power connectors are attached.

6. Test the monitor and PC separately

- Connect the monitor to another working computer.
- If possible, connect another known-good monitor to the affected PC.
- This helps determine whether the problem is the monitor or the computer.

7. Check Windows display settings

- If Windows starts but the display is blank, press Windows + P and select PC screen only or Duplicate.
- Check the display resolution and refresh rate if the screen becomes available.

8. Check RAM and POST

- If there is still no display during startup, power off the PC and reseal the RAM.
- Listen for POST beep codes or check motherboard diagnostic LEDs for VGA, DRAM, or CPU errors.

9. Check the GPU

- Inspect the GPU for dust, damage, or loose connections.
- Reseat it if necessary and test with integrated graphics or a known-good GPU when available.

3) A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

- **Check the mouse itself**

- Make sure the mouse is switched **ON**.
- Replace or recharge the batteries.
- Check whether the optical sensor is clean.
- If the mouse has a USB receiver, make sure it is securely connected.
- **Purpose:** Determines whether the mouse has a power or physical problem.

- **Check the wireless connection**

- Remove and reconnect the USB receiver.
- If the mouse has a pairing button, pair it with the receiver again.
- Keep the mouse close to the receiver while testing.
- **Purpose:** Checks whether the wireless connection has been lost.

- **Test another USB port**

- Move the receiver to another working USB port.
- Wait for Windows to detect the device.
- **Purpose:** If the mouse works in another port, the original USB port may be faulty or disabled.

- **Test the mouse on another computer**

- Connect the wireless receiver to another known-working computer.
- **Purpose:** If the mouse still does not work, the mouse/receiver is likely faulty. If it works, the original PC is more likely to have a driver or settings problem.

- **Check Windows Device Manager**

- Open **Device Manager** → **Mice and other pointing devices**.
- Look for warning symbols or disabled devices.
- Re-enable the device or update/reinstall its driver if necessary.
- **Purpose:** Checks whether Windows is correctly detecting and managing the mouse.

- **Check system settings**

- Open **Settings** → **Bluetooth & devices** → **Mouse**.
- Verify that the mouse is detected and that pointer settings are normal.
- **Purpose:** Rules out incorrect or disabled mouse configuration.

- **Restart and confirm the fix**

- Restart the computer and test the mouse again.
- If the mouse works after changing the USB port or reinstalling the driver, the cause has been identified.